Plaid v2 — Cut Plan

Two end-grain plaid cutting boards built from cherry, walnut, and brown maple.

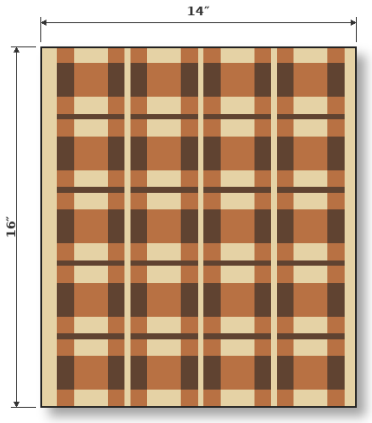
Cut list for the finished design, in build order.

1. Warnings

#	Lvl	Type	Message
1	WARNING	Small Margin	Board BC2A leaves only ⅛″ margin (< ¼″) — too tight; prone to under-supplying strips if the board runs narrow.
2	WARNING	Zero Offcut	30 strip(s) crosscut to an exact fit with 0″ offcut — no margin for cutting error in the crosscut phase: BM1A ×5, BM1B ×7, BC1A ×5, BC2A ×5, BW1A ×3, BW2A ×5. A cut even slightly long, or kerf drift, will make the last piece short. Consider longer blanks or fewer pieces per strip.
3	WARNING	Planer Gang Too Small	There aren't enough 3″ strips to gang together to safely pass through the planer without tipping over. You might need sacrificial wood to tape onto it in order to prevent movement.
4	NOTE	No Spare Slices	No spare slices for: S-L, S-R (supply exactly meets demand). Consider slightly larger/more panels so a bad slice does not cause a shortfall.

2. Finished board pattern

2 finished boards. Final size 14″ × 16″ × 1¼″.



3. Volume Sanity Check

Review the per-species volumes against inventory.

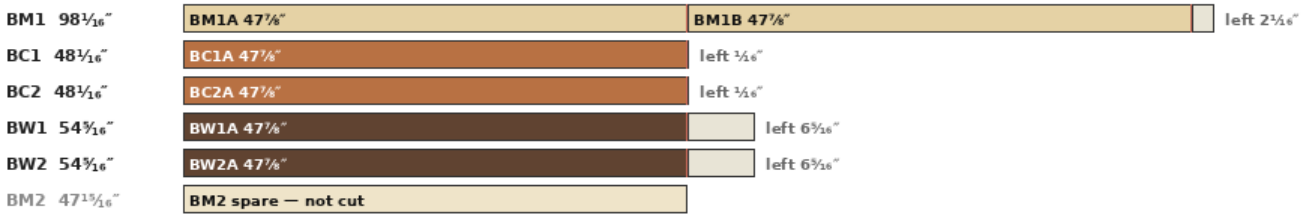
Species	Available	Required	Volume Offset
maple	6.9 BF	1.4 BF + 1.0 BF loss = 2.5 BF	+4.4 BF (277%)
cherry	3.7 BF	1.6 BF + 1.0 BF loss = 2.6 BF	+1.1 BF (144%)
walnut	3.8 BF	1.0 BF + 0.7 BF loss = 1.7 BF	+2.0 BF (218%)

4. Crosscut Blanks

This plan involves cutting blanks from the inventory — review the inventory and the planned cuts.

Raw board	Species	Raw length	Blanks (id = length)	Leftover
BM1	maple	98 1/16″	BM1A = 47 7/8″; BM1B = 47 7/8″	2 1/16″
BC1	cherry	48 1/16″	BC1A = 47 7/8″	1/16″
BC2	cherry	48 1/16″	BC2A = 47 7/8″	1/16″
BW1	walnut	54 5/16″	BW1A = 47 7/8″	6 5/16″
BW2	walnut	54 5/16″	BW2A = 47 7/8″	6 5/16″
BM2	maple	47 15/16″	— (untouched spare)	—

Crosscut Diagram



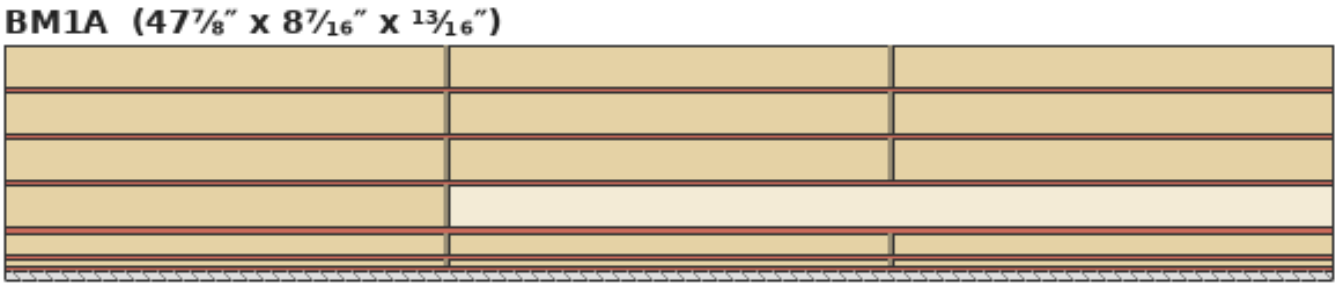
5. Initial board rips (the cut list)

Review the rip list.

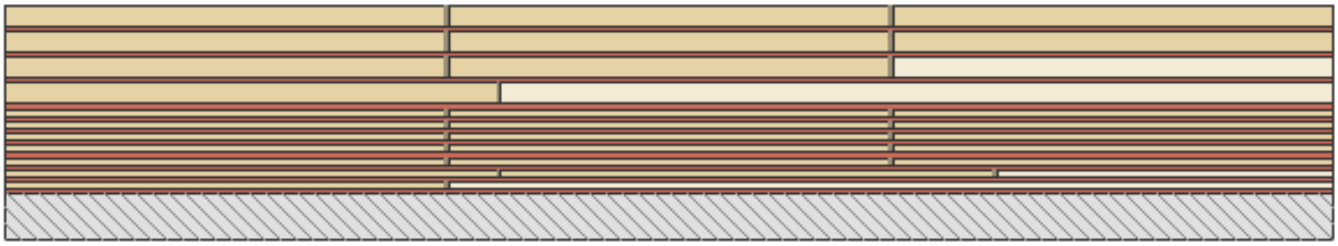
Board	Origin	Species	Length	Rips (width × count)	Used	Loss	Margin
BM1A @ 8 7/16″	BM1	maple	47 7/8″	1 1/2″ ×4 · 3/4″ ×1 · 1/4″ ×1	7″	1 1/8″	5/16″
BM1B @ 8 7/16″	BM1	maple	47 7/8″	3/4″ ×4 · 1/4″ ×7	4 3/4″	2 1/16″	1 5/8″
BC1A @ 7 3/16″	BC1	cherry	47 7/8″	1 1/2″ ×3 · 3/4″ ×2	6″	15/16″	1/4″
BC2A @ 6 1/2″	BC2	cherry	47 7/8″	1 1/2″ ×1 · 3/4″ ×5	5 1/4″	1 1/8″	1/8″
BW1A @ 6 1/4″	BW1	walnut	47 7/8″	3″ ×1 · 3/4″ ×3	5 1/4″	3/4″	1/4″
BW2A @ 6 1/16″	BW2	walnut	47 7/8″	3/4″ ×5	3 3/4″	15/16″	1 3/8″

Board Rip Diagram

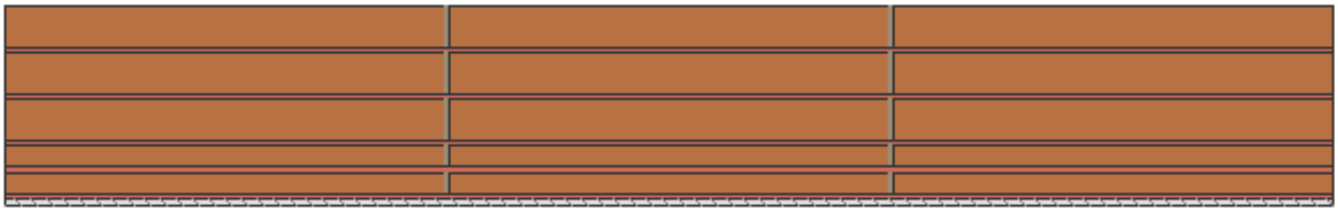
Review the board rip diagrams against the rip list.



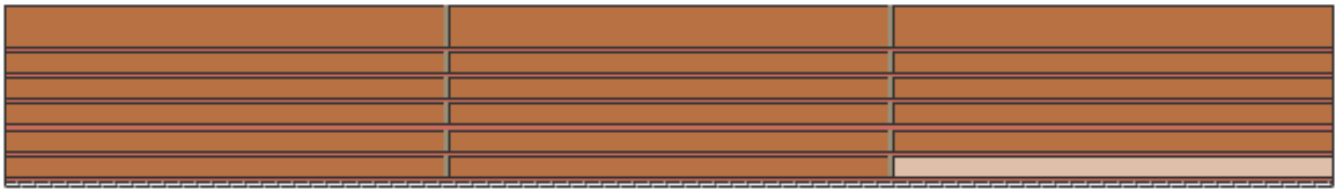
BM1B (47⅞″ x 8⅞₁₆″ x ¹³⁄₁₆″)



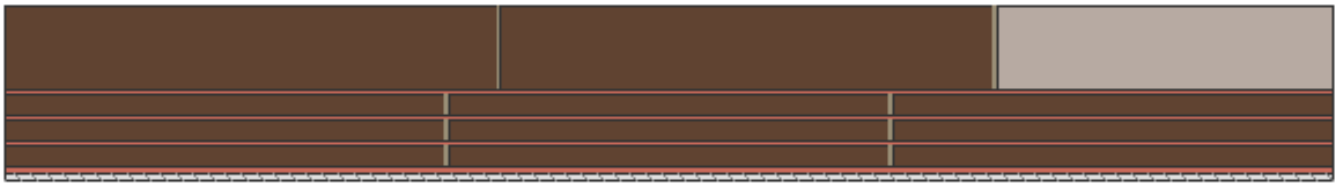
BC1A (47⅞″ x 7³⁄₁₆″ x ¹³⁄₁₆″)



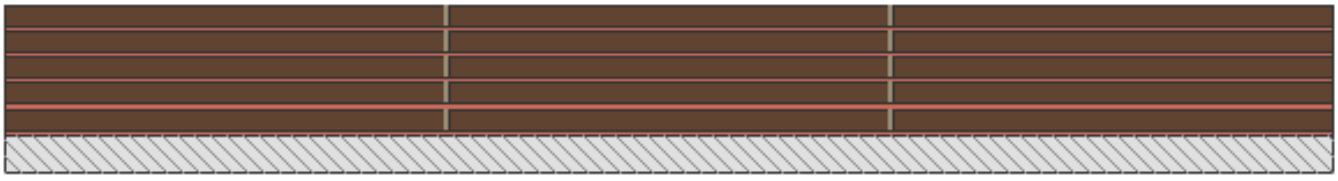
BC2A (47⅞″ x 6½″ x ¹³⁄₁₆″)



BW1A (47⅞″ x 6¼″ x ¹³⁄₁₆″)



BW2A (47⅞″ x 6₁⁄₁₆″ x ¹³⁄₁₆″)



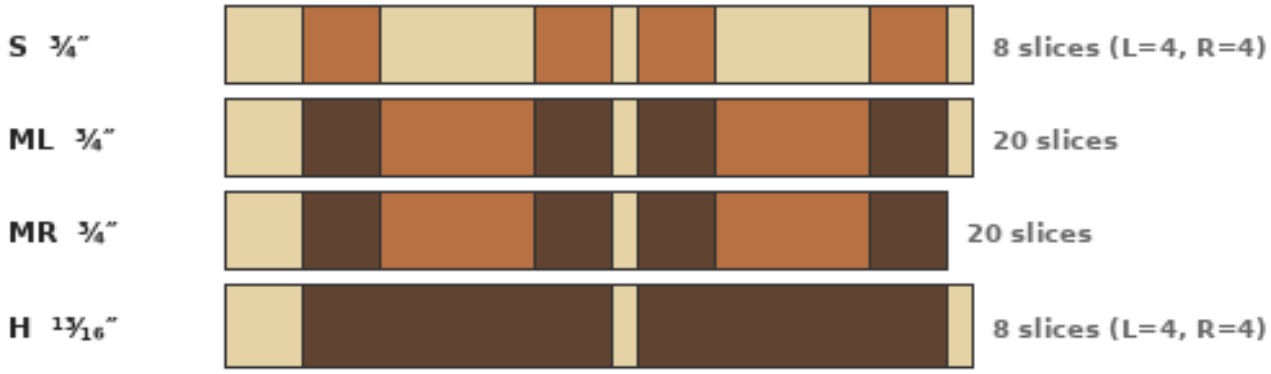
6. Panels overview

Panel	Qty	Length	Slices	Height	Ripped	Sequence (outer → in)
S	5	15⅞″	8 (L:4 + R:4)	¾″	—	Me · C2 · M1 · C2 · Md · C2 · M1 · C2 · Md
ML	3	15⅞″	8	¾″	—	Me · W2 · C1 · W2 · Md · W2 · C1 · W2 · Md

Panel	Qty	Length	Slices	Height	Ripped	Sequence (outer → in)
MR	3	15⅞″	8	¾″	—	Me · W2 · C1 · W2 · Md · W2 · C1 · W2
H	1	17¾″	10 (L:5 + R:5)	¼″	2× (¼″)	Me · W1 · Md · W1 · Md

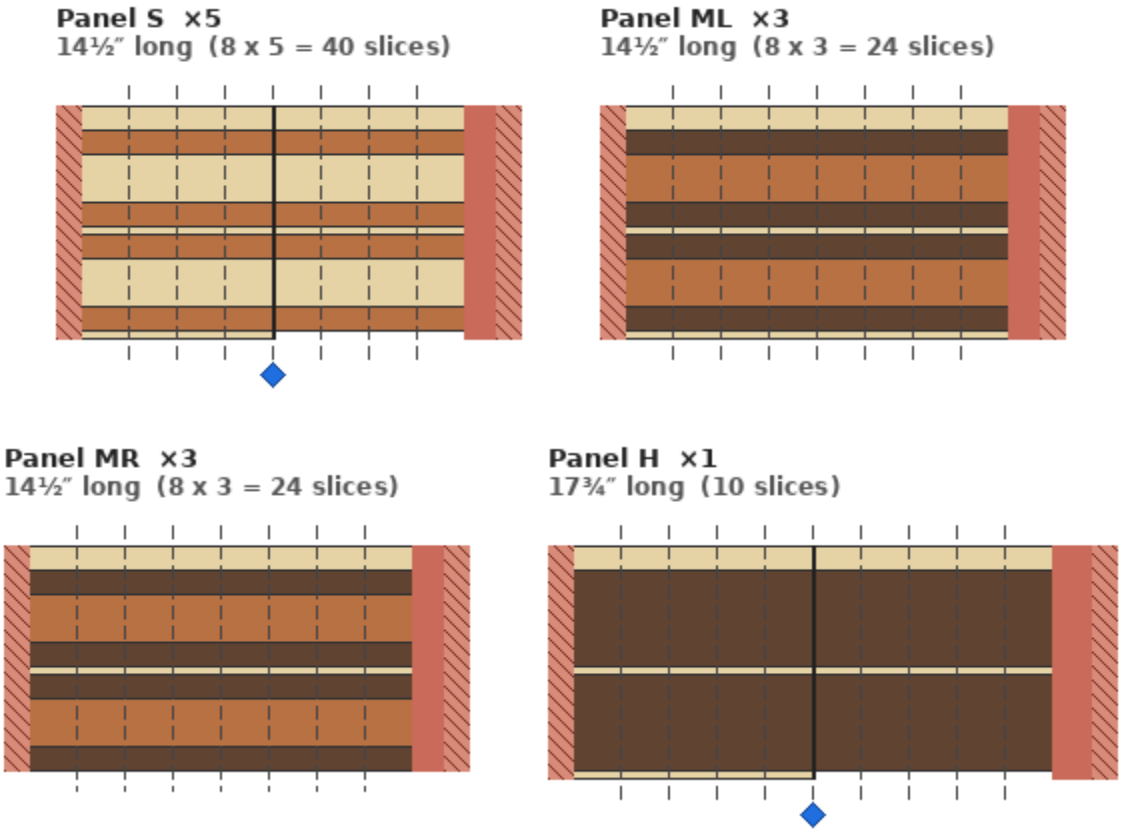
Panel End Grain Diagram

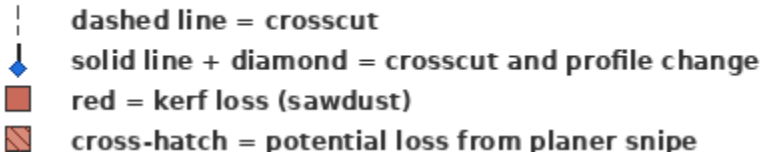
Review the panel end-grain diagrams.



Panel Face Diagram

Review the panel face diagrams.





Slice Fulfillment

Review slice demand, supply, and offset.

Slice	Demand	Supplied	Supply Offset
S-L	10 × 2 = 20	20	0
S-R	10 × 2 = 20	20	0
ML	10 × 2 = 20	24	+4
MR	10 × 2 = 20	24	+4
H-L	4 × 2 = 8	5 × 2 = 10	+2
H-R	4 × 2 = 8	5 × 2 = 10	+2

7. Components

Review component demand, fulfillment, and shortfall.

Component	Species	Width	Length	Demand	Fulfilled	Shortfall
C1	cherry	1½″	15⅞″	12	12	0
C2	cherry	¾″	15⅞″	20	20	0
M1	maple	1½″	15⅞″	10	10	0
Md	maple	¼″	17¾″	2	2	0
Md	maple	¼″	15⅞″	19	19	0
Me	maple	¾″	17¾″	1	1	0
Me	maple	¾″	15⅞″	11	11	0
W1	walnut	3″	17¾″	2	2	0
W2	walnut	¾″	15⅞″	24	24	0

8. Panels (profiles and sequence)

Review the panels and their quantities.

Panel	Qty	Length	Profiles (slices)	Component sequence
S	5	15⅞″	L: 4, R: 4	Me · C2 · M1 · C2 · Md · C2 · M1 · C2 · Md
ML	3	15⅞″	single: 8	Me · W2 · C1 · W2 · Md · W2 · C1 · W2 · Md
MR	3	15⅞″	single: 8	Me · W2 · C1 · W2 · Md · W2 · C1 · W2
H	1	17¾″	L: 5, R: 5	Me · W1 · Md · W1 · Md